

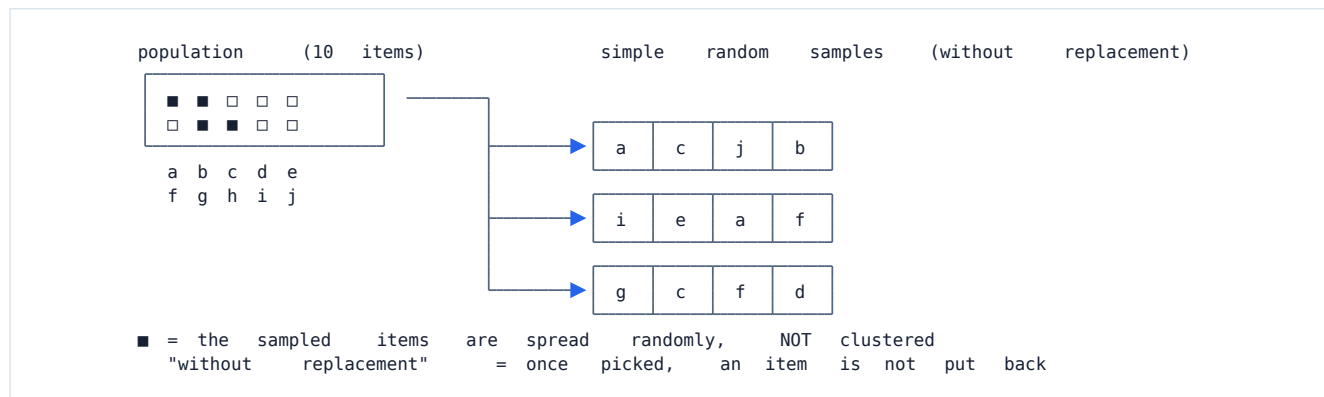
08 SAMPLING AND THE CENTRAL LIMIT THEOREM (lms-standard-error-clt, 19 pages, lectures 17-18)

1. Population vs sample (p001-p002)

POPULATION (N)	SAMPLE (n)
the ENTIRE group you want	a smaller subset, collected
to draw conclusions about	to observe and analyse
described by PARAMETERS	described by STATISTICS
μ, σ	$\bar{X}, s$
often too large / costly	must be representative and
/ impossible to measure	randomly selected
e.g. all 500,000 bulbs	e.g. 100 randomly chosen

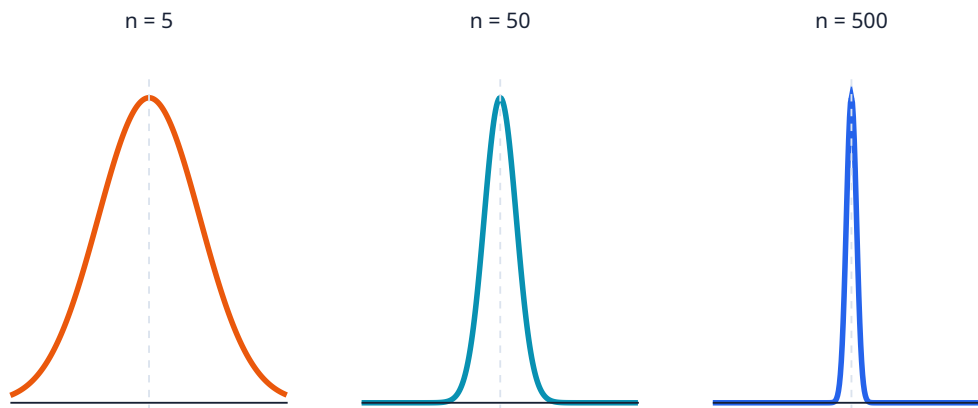
PARAMETER vs STATISTIC - the words the exam checks
 PARAMETER = a number about the POPULATION (μ, σ, p)
 STATISTIC = a number computed FROM THE SAMPLE ($\bar{X}, s, \hat{p}$)
 The parameter is UNKNOWN (that is why we sample).
 The statistic is KNOWN (we compute it).

2. Simple random sampling, the picture (p003)



3. Standard error (p004)

Sampling distributions narrow as n grows



centre stays at μ ; widths use the exact $1 / \sqrt{n}$ ratio

Sampling distributions narrow at the exact square-root rate

STANDARD ERROR measures the variability of a STATISTIC (usually the sample mean) across multiple hypothetical samples. It quantifies how much the estimate deviates from the true parameter.

$$SE = \sigma / \sqrt{n}$$

KEY: as n increases, SE DECREASES.

Larger samples $\rightarrow$ more precise estimates.

the shrinking spread:

n small

 wide SE
 (imprecise)

n medium

 narrower

n large

 very narrow SE
 (precise)

4. WORKED: lightbulbs (p005)

$\sigma = 100$ hours.

Case 1: $n = 25$ $SE = 100/\sqrt{25} = 100/5 = 20$ hours

Case 2: $n = 100$ $SE = 100/\sqrt{100} = 100/10 = 10$ hours

THE SQRT RULE, memorise it:

QUADRUPLING n HALVES the SE

25 $\rightarrow$ 100 is $\times 4$, so SE goes 20 $\rightarrow$ 10 (halved)

this is the answer to every "how large a sample" question

5. The Central Limit Theorem (p007-p010)

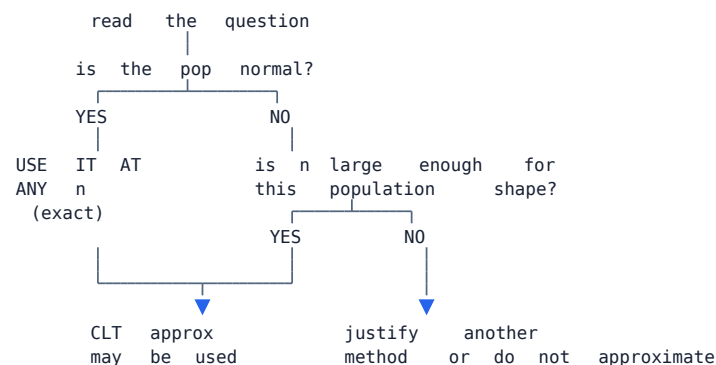
"If you take sufficiently large samples from a population, the distribution of the SAMPLE MEANS will follow a normal distribution, REGARDLESS of the population's underlying shape (even if heavily skewed, uniform, or bimodal)."

THREE RULES:		
1. CENTRE	mean of the sample means = μ	$(\mu_{\bar{X}} = \mu)$
2. SPREAD	sd of the sample means = $\sigma/\sqrt{n}$	$(\sigma_{\bar{X}} = \sigma/\sqrt{n})$
3. SHAPE	as n increases the shape becomes normal;	
$n \geq 30$ is the course rule of thumb		

6. THE THREE CASES - the applicability conditions (p008)

1. NON-NORMAL population, large n
-> sample means may be approximately Normal(μ , $\sigma/\sqrt{n}$)
 $n \geq 30$ is the course heuristic, not a theorem cutoff
2. $n \leq 30$ AND the population IS normal
-> sample means are EXACTLY Normal(μ , $\sigma/\sqrt{n}$)
3. Small n AND the population is NOT normal
-> do not assume a normal approximation without evidence

the decision tree:



7. The visual proof (p010a-p010b)

the 3x4 grid the slide shows:

	NORMAL	UNIFORM	SKEWED	RANDOM
POPULN				
n = 10				
n = 30				

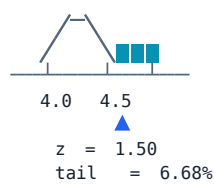
^ all four columns look NARROW BELLS at n=30
that is the CLT: shape washed out, spread shrunk

8. WORKED: ATM wait times (p011-p012)

heavily right-skewed wait, $\mu = 4$ min, $\sigma = 2$ min, $n = 36$.
 $P(\text{average wait} > 4.5 \text{ min})?$

STEP 1	$n=36 \geq 30$, so the sample mean is normal (even though waits are skewed)		
STEP 2	$SE = 2/\sqrt{36} = 2/6 = 0.333$		
STEP 3	$Z = (X - \mu)/SE = (4.5 - 4)/0.333 = 0.5/0.333 = 1.50$		
STEP 4	upper tail at $z=1.50$	->	6.68 %

the picture:



9. WORKED: impurity (p013-p014) - the ERRATA one

$\mu = 4.0$ g, $\sigma = 1.5$ g, $n = 50$.
 $P(3.5 < \bar{X} < 3.8)?$

$SE = 1.5/\sqrt{50} = 0.212132$
 $Z1 = (3.5 - 4)/0.212132 = -2.357 \rightarrow -2.36$
 $Z2 = (3.8 - 4)/0.212132 = -0.943 \rightarrow -0.94$
 $P = P(0 < Z < 2.36) - P(0 < Z < 0.94)$
 $= 0.4909 - 0.3264 = 0.1645$

ERRATA 6: the slide PRINTS $Z2 = -0.4$, which is WRONG.
 The correct value is -0.94 .
 The slide's final 0.1644 is also wrong: $0.4909 - 0.3264$ equals 0.1645 using rounded table areas. Using unrounded z values gives about 0.1637 . WRITE $Z2 = -0.94$ and state which precision convention you used.

10. WORKED: LED bulbs (p015-p016)

"heavily skewed", $\mu = 50000$, $\sigma = 8000$, $n = 64$.
 $P(X < 48000)?$

$SE = 8000/\sqrt{64} = 8000/8 = 1000$	
Z	$= (48000 - 50000)/1000 = -2.00$
P	$= 0.0228$

note the SAME $-2.00 \rightarrow 0.0228$ as the normal example.
 It is the same table read, just reached through SE instead of σ .

11. WORKED: machine life, small n (p017-p018)

"the lives follow the NORMAL distribution", $\mu = 7$, $\sigma = 1$, $n = 9$.
 $P(6.4 < X < 7.2)$?

$$SE = 1/\sqrt{9} = 1/3$$

$$Z_1 = (6.4 - 7)/(1/3) = -1.80$$

$$Z_2 = (7.2 - 7)/(1/3) = 0.60$$

$$P = 0.4641 + 0.2257 = 0.6898$$

n = 9 IS BELOW 30, BUT THE QUESTION SAYS THE POPULATION IS NORMAL. That is case 2: the sample mean is EXACTLY normal. So we can still use the table. The word "normal" in the question gives an exact result, so no $n \geq 30$ heuristic is needed.

the two tails added:



THE SAMPLING/CLT KIT

SE of a mean	$\sigma/\sqrt{n}$ (or $s/\sqrt{n}$)
SE of a proportion	$\sqrt{p(1-p)/n}$
Z for a mean	$(\bar{X} - \mu) / (\sigma/\sqrt{n})$
Z for a single obs	$(X - \mu) / \sigma$ <- NOT the same!
n >= 30	course heuristic for CLT approx
n < 30, normal pop	exact normal
n < 30, non-normal	normal approx needs justification

THE SINGLE COMMONEST MISTAKE IN THIS TOPIC:
 using σ in the denominator for a question about the AVERAGE
 of n items. An average ALWAYS divides by $\sigma/\sqrt{n}$.
 one bulb's life -> divide by σ
 the MEAN of 25 bulbs -> divide by $\sigma/\sqrt{25} = \sigma/5$